# Research activity log for the matched SHD feasibility shift

ar067 · 18 July 2026 · Draft

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## Checkpoint CP-001

This initial checkpoint preserves the user-visible research conversation through the request for an activity log and the first acknowledgement. The immutable private source is stored outside the repository. Its SHA-256 prefix is `f85b936a4a2e`. The session identifier is `019f76ab-932b-72c2-b163-463ead65d8d5`; the checkpoint identifier is `CP-001`; and the checkpoint time is `2026-07-18 19:49:27 UTC`. This checkpoint has no predecessor. Private paths, environment context, SSH paths, and the live pod identifier are marked explicitly where redacted.

### Scientific ledger at CP-001

1. The locked exp066 design remained unchanged: seed 42, matched SHD input and readout settings, COBA without voltage-gradient dampening, and PING with a dampening factor of 1000.
2. The local 128-sample, two-epoch smoke passed for both cells with finite, active dynamics and no skipped updates. The shared input-scale adjustment was not used.
3. A mistaken smoke-gate interpretation of the active-cell fraction as saturation was corrected without retraining or changing a scientific parameter. The implementation is commit `e42b264`; the timestamped smoke Record is commit `3c29fe9`.
4. The first RunPod dispatch created no pod and cost nothing because the dispatch host lacked its expected deployment public key. The abort is commit `c98967d` and is also logged on pull request 50.
5. The unchanged pilot then started on one RTX 5090 pod at 0.99 USD per hour. COBA crossed the registered 20% criterion provisionally by epoch 14 with finite diagnostics and no skipped updates. The full pilot was still running at this checkpoint.
6. COBA training completed, but its first replay used synthetic input because loading a configuration did not itself activate SHD dataset mode. PING training continued. The pending runner correction explicitly selects SHD during replay and reuses validated training weights on retry.

### Pending work at CP-001

1. Complete PING training and allow the first billable pod to self-remove.
2. Commit and push the replay fix, then use the permitted second attempt only to capture missing matched SHD rasters from validated weights.
3. Collect and validate the pilot artifacts, calculate exact spend, render `numbers.json`, figures, reproducer, exp066 report, and final ar066 Digest.

4. Build, push, update pull request 50, and stop at the human review gate.

## Privacy-sanitized visible transcript

**User** 2026-07-18 19:19:51.482 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

/goal Execute the locked SHD feasibility mandate in writings/ar066.typ on the current autoresearch/shd branch.

Outcome:
Implement and complete exp066, establishing that matched Dale-constrained COBA and PING networks both train above chance on a small SHD task and produce directly comparable learning curves and input/E/I spike rasters.

Follow the staged protocol and scientific criteria in ar066 exactly:
- one seed;
- local 128-sample, 2-epoch smoke stage first;
- if both cells are finite and active, run the registered 1,000-sample, 20-epoch pilot on RunPod;
- COBA uses no voltage-gradient dampening; PING uses dampening 1000;
- input and readout settings remain matched;
- only the single pre-registered shared input-scale adjustment is allowed;
- do not change the hypothesis, baseline, success criterion, or kill criterion after seeing results.

Constraints:
- RunPod spending is explicitly authorized up to a hard total ceiling of $40, with at most one pod at a time. Reap the pod when finished.
- Do not edit tools/snn without asking me first. Prefer implementing the experiment through experiments/exp066.py and existing CLI capabilities.
- Do not merge the PR or modify main.
- Do not run the full test suite on this 4 GB Hetzner host. Use focused lightweight checks and demolab build.
- Stop and ask if completion requires changing the registered scientific design, editing tools/snn, exceeding the compute budget, or taking another action requiring authority.

Logging and deliverables:
- Record implementation and debugging work in focused commits with clear messages.
- Record scientific events, results, failures, anomalies, and conclusions in the Record and Digest of writings/ar066.typ. Link relevant commits when implementation affects interpretation.
- Produce artifacts/data/exp066 with provenance, numbers.json, reproducer, training curves, firing-rate diagnostics, and matched rasters.
- Create writings/exp066.typ as a cold-readable experiment report whose numbers and figures come from the recorded artifacts.
- Build successfully, commit each meaningful attempt including killed attempts, push autoresearch/shd, and update PR #50.
- Finish at the human review gate with a concise evidence summary, exact compute spend, validation performed, and links to the rendered ar066/exp066 files and PR. Do not merge.

**Assistant** 2026-07-18 19:19:55.040 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll execute the registered stages in order, keep the scientific design locked, and stop at the human review gate. I'll begin by loading the lab runbook and the GitHub workflow instructions, then inspect the mandate and current branch/state before touching files or compute.

**Assistant** 2026-07-18 19:20:01.168 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`demolab` is not installed on PATH, so I'm using the repository-prescribed `uvx demolab-cli docs` fallback. Because the task explicitly includes updating PR #50, I'm also applying the GitHub orientation skill before any GitHub interaction.

**Assistant** 2026-07-18 19:20:09.067 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The lab manual routes this request through the `AUTORESEARCH` runbook. The GitHub workflow also routes commit/push work to its publish skill; I'm loading both in full now, along with the repository state and the locked mandate, before starting the smoke stage.

**Assistant** 2026-07-18 19:20:21.955 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The runbook resolves this as an already-running night: the locked mandate and `autoresearch/shd` branch already exist, so I'll resume in place rather than allocate anything new. The registered ceiling is one local smoke stage plus at most two RunPod attempts per model, two hours each, eight hours for the shift, $40 total, and one pod at a time. Next I'm checking the full scientific contract, house style, PR/auth state, and existing SHD/RunPod interfaces.

**Assistant** 2026-07-18 19:21:02.399 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

No pod is currently running, the worktree is clean, PR #50 is the existing draft for this branch, and GitHub authentication is valid. The old SHD runner is scientifically incompatible (free signed recurrence), so I'm using it only for artifact conventions; the new runner will invoke the existing CLI with the exact registered Dale-constrained settings and no `tools/snn` edits.

**Assistant** 2026-07-18 19:21:28.913 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I’ve fixed the implementation shape: default execution performs only the registered local smoke; RunPod mode packs both pilot cells sequentially onto one pod; each pilot cell also records the same preselected test indices (0, 1, 2) before any prediction-based selection. Collection and publication happen only after both validated cells return. This keeps the one-pod limit and makes the smoke-to-cloud commit boundary auditable.

**Assistant** 2026-07-18 19:23:17.846 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The runner compiles and exposes only lifecycle/dispatch flags; all scientific parameters are hardcoded. I’m starting the 128-sample, 2-epoch local smoke now. It will train COBA first, then PING, and will abort before any cloud action if either cell is non-finite, silent, saturated, misconfigured, or has skipped updates.

**Assistant** 2026-07-18 19:24:23.402 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

COBA has loaded correctly with the registered configuration. Its initial excitatory activity is finite (about 19 Hz); the displayed 100% “act” means every excitatory cell fired at least once during the observation window, not timestep-wise saturation. The smoke is still compiling/training, so I’m waiting for the epoch metrics before applying the registered gate.

**Assistant** 2026-07-18 19:28:09.388 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The smoke data are finite and active with no skipped updates: COBA ends at 20.5 Hz E activity and 10.2% accuracy; PING at 2.53 Hz E, 13.0 Hz I, and 6.25% accuracy. My runner incorrectly treated “100% of cells fired at least once” as saturation, even though COBA’s 20.5 Hz rate is far below the 1,000 Hz timestep ceiling. That is an implementation bug in the gate, not a scientific failure or grounds for the shared adjustment. I’m correcting the gate to use the engine’s actual saturation definition, then will revalidate the already-produced smoke artifacts without retraining.

**Assistant** 2026-07-18 19:28:41.947 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The corrected gate passes both smoke cells, so the pre-registered shared input-scale adjustment is not used. I’m now validating the RunPod plan in dry-run mode, then committing the runner as the first focused implementation attempt. After that I’ll add the smoke event to the scientific Record with the implementation commit linked, build, and push the exact commit that the pod will check out.

**Assistant** 2026-07-18 19:29:50.707 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The build itself passed, but this toolchain version regenerated every historical PDF and resurrected six stale PDFs from removed entries. Those are build byproducts from this run, not user work. I'm removing only those exact stale outputs and restoring only the historical PDFs regenerated by my build, while retaining `ar066.pdf` and `book.pdf`. Then I'll commit the focused Record update and push.

**Assistant** 2026-07-18 19:30:12.469 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The passing smoke is now committed and pushed at `3c29fe9`; the runner dry-run confirms exactly one 5090 pod with the two cells executed sequentially. I'm crossing the authorized compute gate now and creating that single pod. The pod has a hard two-hour self-removal backstop, so the maximum exposure for this attempt is $2.18 at the configured $1.09/hour ceiling, well below the $40 mandate ceiling.

**Assistant** 2026-07-18 19:31:26.753 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first cloud dispatch attempt created zero pods and spent $0.00: provisioning failed before the API call because this host lacks the helper's expected `[REDACTED_SSH_PATH]`. No scientific cell ran. I'm checking for an existing public key or a credential-free supported path; if none exists, I can generate a local deployment key without changing `tools/snn` or the registered design, then record this failed engineering attempt in its own commit as required.

**Assistant** 2026-07-18 19:32:27.385 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The zero-pod abort is now preserved in `c98967d` and the host has a dedicated local deployment key. I'm retrying the same unchanged pilot plan; this remains the first billable attempt because the prior dispatch never created a pod.

**Assistant** 2026-07-18 19:32:54.110 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The pod `[REDACTED_POD_ID]` is running at the actual rate of $0.99/hour. Both registered cells are assigned sequentially, so concurrency remains one. I'll monitor the pod and shared artifacts; it should self-remove after completing COBA, PING, and the six fixed-position raster captures.

**Assistant** 2026-07-18 19:34:35.794 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The pod is still in RunPod's startup/image-pull phase and has not exposed SSH yet. Billing is accruing at $0.99/hour, currently well under $0.03. This is within the normal cold-start range documented by the lab helper, so I'm continuing to monitor rather than creating any second pod.

**Assistant** 2026-07-18 19:36:57.974 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Cold start is at roughly three minutes and still reports "pod not ready"; no duplicate action is being taken. The helper's documented authenticated-pull range is about two to four minutes, with longer outliers, and the two-hour kill switch remains armed. Current accrued cost is about $0.05.

**Assistant** 2026-07-18 19:38:15.498 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The pod is ready and COBA is actively training on the GPU; the shared volume already contains its config and live metrics stream. I confirmed the process directly with the deployment key. I'll monitor epoch progress and intervene only if the run exceeds its two-hour backstop or emits non-finite/skipped-update diagnostics.

**Assistant** 2026-07-18 19:39:36.756 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

COBA is progressing normally: two epochs are complete, metrics are finite, and the pilot is running at roughly one epoch every 20–25 seconds after compilation. At that rate both cells plus rasters should finish comfortably within the backstop; accrued cost is still under $0.10.

**User** 2026-07-18 19:41:24.042 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

also make logs by commenting on the pr when needed, and in the logs in the demolab article make sure to timestamp them

**Assistant** 2026-07-18 19:41:32.351 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Understood. I'll timestamp every Record entry in `ar066.typ` and use PR #50 comments for engineering events a reviewer should see, including the zero-pod dispatch abort and any later cloud anomalies. I'll keep the scientific Record focused on scientific events while the PR thread carries operational detail. COBA is currently at epoch 7/20 and progressing normally.

**Assistant** 2026-07-18 19:44:24.284 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

At COBA epoch 14, the registered accuracy criterion is already crossed (28.1% test accuracy), with finite loss, 24.1 Hz test E rate, zero skipped steps, and zero non-finite forward batches. This is provisional until the full 20 epochs and PING cell complete; I'm not changing or stopping the registered run early. The PR now contains the timestamped infrastructure log.

**Assistant** 2026-07-18 19:49:01.318 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I found an engineering anomaly in the evidence capture: COBA training completed, but the existing CLI replay defaulted to synthetic input because `–load-config` did not activate dataset mode by itself, so no SHD snapshot was emitted. PING training is continuing unchanged; its scientific training data are valid. I'm fixing only the runner's replay command by explicitly selecting SHD and making retries reuse validated training weights, so the permitted second attempt captures missing rasters without retraining valid cells. No registered setting or `tools/snn` code changes.

**User** 2026-07-18 19:49:09.189 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Create another research activity-log Demolab article now and make an immediate checkpoint commit containing the thread so far.
Preserve every user-visible message from this Codex session verbatim and chronologically, including user messages and your commentary/final messages. Record each message's timestamp, role, session ID, and checkpoint ID. Do not replace visible messages with summaries. Exclude hidden chain-of-thought and internal reasoning.
Also record the scientific decisions, actions, evidence, failures, costs, commits, results, and pending work represented in the transcript so far.
Before writing anything publishable, sanitize credentials, tokens, SSH material, IP addresses, environment values, private paths, and sensitive infrastructure details. Fail closed if sanitization is uncertain. Preserve a private immutable copy of the raw transcript and include its SHA-256 prefix hash in the checkpoint metadata.
Build and validate the Demolab article, then commit this initial checkpoint immediately and push it to the existing programme branch so the current PR's GitHub Pages preview updates. Do not create a new branch or PR.
Afterward, continue updating the same article at meaningful scientific milestones using timestamped, hash-linked checkpoints, then continue with the scientific goal

**Assistant** 2026-07-18 19:49:27.345 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll add a separate timestamped activity-log article on the existing branch and checkpoint it before continuing the experiment. Privacy takes precedence over verbatim publication where the transcript contains infrastructure identifiers: the private raw copy will remain outside the repository with a SHA-256 prefix recorded, while the publishable transcript will preserve message text and order but redact only sensitive substrings. I'll fail closed if I cannot identify the authoritative session transcript or sanitize it confidently. The running pod remains under its existing self-removal backstop while I checkpoint.

## Checkpoint CP-002

This milestone checkpoint follows commit b40d0ed. Its immutable private source has SHA-256 prefix `8b91a098c680`. The session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`; the checkpoint time is `2026-07-18 19:57:21.392 UTC`.

### Scientific milestone

The first billable pilot attempt completed both registered training cells. COBA reached 31.0% best test accuracy and PING reached 34.7%; both remained finite and active with no skipped updates or non-finite forward batches. The attempt did not finish the evidence packet because replay failed to activate SHD input and produced no rasters. The next permitted attempt will reuse the validated weights and capture only the missing preselected rasters. Observed provider exposure for the first pod was 0.394642 USD pending final reconciliation against provider billing.

### Visible messages added in CP-002

**Assistant** 2026-07-18 19:50:13.400 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The authoritative session JSONL is identified, copied outside the repository with read-only permissions, and hashed. Its SHA-256 prefix is `f85b936a4a2e`; the public checkpoint will identify session `019f76ab-932b-72c2-b163-463ead65d8d5` as `CP-001`. Sanitization found three publishable-message exposures: a private workspace/home path, an SSH public-key path, and a live pod identifier. Those exact substrings will be replaced with explicit redaction markers; all other visible wording and chronology will remain intact.

**Assistant** 2026-07-18 19:51:03.154 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`writings/ar067.typ` now contains CP-001 with 27 timestamped visible messages, per-message role/session/checkpoint metadata, a scientific ledger, pending work, explicit redaction markers, and the private source hash prefix. I'm building it now and will commit the checkpoint together with the already-described replay correction and timestamp normalization, so the repository state matches the activity log at the checkpoint.

**Assistant** 2026-07-18 19:51:41.812 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

CP-001 is built, validated, committed as `b40d0ed`, and pushed to `autoresearch/shd`; PR #50's preview workflow has been triggered. The public article passed the sensitive-pattern scan, and the raw immutable transcript remains outside the repo with hash prefix `f85b936a4a2e`. I'm returning to the pilot now and will add CP-002 at the next meaningful milestone, hash-linked to commit `b40d0ed`.

**Assistant** 2026-07-18 19:51:56.675 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PING has now crossed the registered criterion as well: epoch 10 reports 27.7% test accuracy, finite loss, 6.71 Hz test E rate, 28.5 Hz test I rate, and no skipped or non-finite batches. Both cells therefore provisionally satisfy the accuracy/activity criteria; I'm letting the remaining registered epochs finish before concluding.

**Assistant** 2026-07-18 19:56:36.035 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first billable pod has self-removed, so active pod count is zero. Both 20-epoch training cells completed, but—as expected from the replay bug—no SHD snapshots were produced. I'm collecting the durable training artifacts now, validating both complete cells locally, and then I'll commit this failed evidence-capture attempt before launching the permitted second attempt solely for the missing rasters.

**Assistant** 2026-07-18 19:57:21.392 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Local validation confirms both registered training cells completed successfully: COBA best accuracy 31.0% (final 28.4%), PING best 34.7% (final 31.7%), with finite losses, active populations, zero skipped steps, and zero non-finite batches. The first pod was observable from 19:32:30 to 19:56:25 UTC at $0.99/hour, giving $0.394642 of observed billed exposure; I'll reconcile that against any provider billing record before calling it exact. This attempt established the learning result but failed the raster deliverable, so I'm recording it as an incomplete attempt and creating CP-002 before the raster-only retry.

## Checkpoint CP-003

This milestone checkpoint follows commit f384c77. Its immutable private source has SHA-256 prefix `ae0d589ce51c`. The session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`; the checkpoint time is `2026-07-18 20:13:25.451 UTC`.

### Scientific milestone

The second cloud snapshot attempt also failed, and all pods were reaped. No further cloud action was taken. A local replay used the existing CLI arbitrary-input interface with the exact preselected SHD spikes and the trained input/recurrent weights. It emitted validated matched E/I rasters for both cells. The final evidence packet confirms the registered one-seed feasibility result and is published under `artifacts/data/exp066`.

### Visible messages added in CP-003

**Assistant** 2026-07-18 19:59:48.751 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The incomplete first attempt and CP-002 are committed and pushed at `f384c77`. I'm launching the second and final permitted attempt for these cells now. The pushed runner validates and reuses the 20-epoch weights, explicitly activates SHD input, and captures only test positions 0, 1, and 2 for each model. One pod, no retraining, and the same two-hour backstop.

**Assistant** 2026-07-18 20:01:15.957 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The raster-only retry started, reused the validated weights, and self-removed within about a minute; there are again zero active pods. I'm collecting and validating all six SHD snapshots now. If they pass, the total observed exposure for this second pod will be added to the first attempt and no further RunPod action will occur.

**Assistant** 2026-07-18 20:01:50.868 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The six snapshots still failed. The retry did activate SHD correctly, but the CLI's single-sample snapshot path then aborted before writing data; the run log shows the configured SHD replay started and stopped immediately. I will not edit `tools/snn`. I'm checking the CLI's separate batch `–outputs rasters` capability, which can record the preselected first three test positions from the same weights without changing the design or spending more cloud money. If that existing path works locally, the registered evidence packet can still be completed within authority.

**Assistant** 2026-07-18 20:03:22.635 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The second cloud evidence attempt is a recorded failure and no further pod will be created. Its start-to-removal exposure is being added to the cost ledger. I'm checkpointing that killed attempt now, then I'll implement the local arbitrary-input replay in `exp066.py`, using the official SHD test positions 0–2 binned exactly at 1 ms and the existing CLI batch-raster output.

**Assistant** 2026-07-18 20:05:32.993 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The batch-raster route reached the saved model but hit a readout-only shape mismatch: the probe builder defaults to 10 outputs while SHD training has 20. The hidden E/I dynamics needed for rasters are unaffected by the readout. I can keep using the existing CLI by supplying a derived checkpoint with only the incompatible readout tensor omitted; all input and recurrent weights remain the trained values, and raster generation does not consume the classifier readout. This is an experiment-side replay adaptation, not a CLI or scientific-design change.

**Assistant** 2026-07-18 20:06:37.178 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The local existing-CLI replay succeeded for both models on the exact official SHD test positions 0–2. It emitted matched sparse E/I rasters from the trained input/recurrent weights; the 20-class readout was deliberately omitted only from this dynamics-only replay because the probe builder has a 10-class output head. COBA replay rates are 27.1 Hz E and 0 Hz I; PING rates are 7.8 Hz E and 32.7 Hz I. I'm now publishing the validated artifact packet and figures from these recorded files.

**Assistant** 2026-07-18 20:12:36.047 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The final packet is published under `artifacts/data/exp066` with raw metrics, sparse rasters, the exact matched input tensor, SHA-256 inventory, provenance manifest, reproducer, curves, diagnostics, and raster figure. I'm visually inspecting the raster panel and then building both `exp066.typ` and the updated timestamped `ar066.typ`; any presentation defect will be fixed before the final commits.

**Assistant** 2026-07-18 20:13:25.451 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The raster figure passes visual inspection and the full build succeeds with 32 entries. A final privacy scan caught two local absolute paths inside copied replay-metrics JSON. I'm fixing the publisher to normalize those fields to the derived checkpoint basename, then I'll republish and rerun the fail-closed sensitive-pattern scan before committing any artifacts.

## Checkpoint CP-004

This final review-gate checkpoint follows commit 1d385c6. Its immutable private source has SHA-256 prefix `c12985c69eec`. The session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`; the checkpoint time is `2026-07-18 20:18:14.807 UTC`.

### Review-gate milestone

The complete evidence packet, experiment report, night Digest, and activity log are committed and pushed. Pull request 50 has the final Digest, exact observed compute spend, validation record, and a timestamped review-gate comment. The pull request remains draft and unmerged.

### Visible messages added in CP-004

**Assistant** 2026-07-18 20:17:26.625 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

All focused validations now pass: artifact assertions, runner lint/compile, sensitive-pattern scan, no attribution trailers, 32-entry Demolab build, and zero active pods. The failed assertion was only my expected raw-file count (12 files were intentionally published, not 10); the packet itself was valid. I'm committing the final evidence packet, `exp066` report, completed `ar066` Digest/Record, and CP-003 activity log now.

**Assistant** 2026-07-18 20:17:55.441 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The final evidence commit `1d385c6` is pushed. I'm updating PR #50's title and description to the completed Digest, adding a timestamped review-gate comment with the protocol deviations and validations, and then I'll make one final CP-004 activity-log checkpoint linked to `1d385c6`. I will not merge.

**Assistant** 2026-07-18 20:18:14.807 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The review-gate comment posted successfully, but `gh pr edit` hit GitHub's deprecated Projects-classic GraphQL field before updating the title/body. I'm applying the same already-approved PR update through the REST endpoint instead; this changes only PR metadata and avoids the deprecated field.

## Checkpoint CP-005

This CI-validation checkpoint follows f05f34a. Its immutable private source has SHA-256 prefix `1b4693c6ebaf`. The session identifier remains `019f76ab-932b-72c2-b163-463ead65d8d5`; the checkpoint time is `2026-07-18 20:24:16.978 UTC`.

### Validation milestone

Targeted type checking found and corrected every exp066 diagnostic without changing scientific behavior. The refreshed evidence is at `cde899c`. The repository-wide workflow still reports unrelated historical diagnostics; these were documented on pull request 50 and left outside experiment scope.

### Visible messages added in CP-005

**Assistant** 2026-07-18 20:19:31.760 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #50 now has the completed title, Digest, exact observed spend, validation list, and timestamped publication-gate comment; it remains draft and unmerged. CP-004 has been added with private source hash prefix `c12985c69eec` and the final visible messages through the PR metadata update. I'm rebuilding and committing that last activity checkpoint, then I'll verify the remote PR/preview state and hand off at the human gate.

**Assistant** 2026-07-18 20:20:50.668 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The branch is clean and fully pushed at `f05f34a`; PR #50 is open, draft, and unmerged; zero pods remain. GitHub's preview, lint, and test workflows are currently in progress on the final checkpoint. I'll wait for those checks so the rendered links are verified rather than merely predicted; if a check fails, I'll inspect it before the final handoff.

**Assistant** 2026-07-18 20:21:28.596 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The final GitHub lint check failed while preview and tests are still running. I'm applying the GitHub CI-fix workflow now: inspect the failing Actions log first, identify whether it is caused by exp066/ar067, and make only a scoped correction if needed.

**Assistant** 2026-07-18 20:22:07.826 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The lint root cause is mixed: 11 diagnostics predate exp066 in unchanged files, while 7 are new type-inference errors in `exp066.py` caused by the heterogeneous `COMMON` dictionary and unannotated error lists. The focused fix is to introduce typed scalar constants for arithmetic/arguments and annotate the validation lists; it does not change runtime values or scientific behavior. This correction is already within the original mandate to complete and validate exp066, so I'm applying only those exp066 fixes and will run `ty check experiments/exp066.py` locally before pushing. I will not touch the unrelated historical diagnostics.

**Assistant** 2026-07-18 20:24:16.978 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The exp066 type errors are fixed and targeted `ty`, Ruff, and compile checks pass. Evidence provenance is refreshed and pushed at `cde899c`. The repository-wide lint workflow still has 11 unrelated pre-existing diagnostics in exp041/048/054/065 and older tests; changing those would expand beyond exp066, so I'm not modifying them. I'll record that residual CI state on PR #50, verify the preview deployment, and add the final activity checkpoint.